
Learning to Maintain Epistemic Possibilities in Discourse

Human Behaviour, Linguistic Input, and Controlled Language-Model Training

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Abstract

Epistemic expressions do more than indicate how certain a speaker is. They may also differ in which possibilities they make available for subsequent thought and discourse. For example, *This might be a lemon* explicitly introduces the possibility that the object is a lemon, whereas *I do not know whether this is a lemon* reports non-resolution of the same issue. The two utterances mention the same proposition and are both compatible with uncertainty, yet they may differ in how readily the lemon possibility is subsequently accessed. This thesis asks whether such a difference can be established behaviourally and, if so, what properties of linguistic input are sufficient for a controlled learner to acquire it. An adult reaction-time study will provide the human behavioural target. A corpus audit will then identify candidate distributional and cross-sentence cues, and matched training-data interventions will test their causal contribution in small language models trained from scratch. The project is designed to connect a substantive question about modal language and cognition with a tightly controlled computational learnability experiment, without treating language models as direct models of children.

1. Motivation and research gap

Formal work on epistemic modality distinguishes possibility from necessity relative to an information state (Kratzer, 1981). Dynamic and inquisitive accounts further show that modal utterances can introduce hypothetical discourse structure and direct attention to particular possibilities (Roberts, 1989; Ciardelli, Groenendijk, & Roelofsen, 2009). These theories motivate a behavioural question that is not settled by truth conditions alone: does the way an unresolved issue is linguistically packaged affect how available a particular alternative remains after the utterance?

Existing studies have investigated the comprehension and development of epistemic force (Ozturk & Papafragou, 2015), the distributional cues to epistemic readings in child-directed speech (van Dooren et al., 2022), and the sensitivity of pretrained language models to epistemic expressions (Li, Vrazitulis, & Schlangen, 2025; Du et al., 2026). Controlled training methods have also shown how removing or modifying evidence in a corpus can reveal what neural learners generalize from indirect input (Patil et al., 2024; Yang, Getz, & Wilcox, 2026). The nearest work found does not, however, combine a calibrated human measure of the downstream availability of a particular epistemic possibility with causal manipulation of the discourse-level input from which that behaviour might be learned.

2. Research questions and hypotheses

RQ1 · Human behaviour. When lexical content, uncertainty, timing, and discourse context are controlled, do epistemic expressions differ in how readily a specific possibility is accessed after the utterance?

RQ2 · Input evidence. Which distributions or cross-sentence relations in developmentally relevant language input could support the acquisition of this behavioural contrast?

RQ3 · Computational learnability. For a fixed learner, does preserving, removing, disrupting, or restoring these input relations change whether the target behaviour is acquired and generalized?

The primary behavioural hypothesis is an interaction rather than a general speed advantage. Relative to closely matched expressions, *might p* is predicted to facilitate responses specifically to material related to *p*, and this advantage should weaken when later evidence explicitly excludes *p*. The learning hypothesis is that disrupting the association between an epistemic introducer and the later maintenance, use, or withdrawal of *p* will selectively weaken the corresponding model behaviour; restoring that association should restore the effect.

3. Proposed studies

3.1 Adult behavioural calibration

The human component will test the post-utterance accessibility of a named possibility. Participants will hear or read short, tightly matched contexts containing four schematic conditions:

1. **Epistemic possibility:** *This might be a lemon.*
2. **Explicit non-resolution:** *I do not know whether this is a lemon.*
3. **Lexical mention control:** the word *lemon* occurs without introducing the proposition that this object is a lemon.
4. **Evidence-based withdrawal:** *This might be a lemon*, followed by evidence that it is not.

The principal readout will be a speeded picture-verification or target-detection task of the kind previously suggested by Charlotte. Probe relatedness and probe delay will be manipulated so that a content-specific and persistent accessibility effect can be distinguished from general response speed and short-lived lexical priming. Accuracy will provide a secondary measure. Materials will first receive a small independent naturalness and interpretation calibration; the confirmatory analysis will use mixed-effects models with participants and items as crossed random factors.

This measure has a deliberately narrow interpretation. Faster responses would support greater cognitive accessibility of the target concept or possibility; they would not by themselves establish a persistent neural representation or a particular formal-semantic mechanism. A short discourse-continuation calibration will therefore be used to confirm that the materials also express the intended contrast between maintaining and withdrawing *p*, without turning the MA into a second full human experiment.

3.2 Input audit

The corpus component will examine child-directed and human-scale corpora, beginning with CHILDES and BabyLM-style data. The unit of analysis will not be modal-word frequency alone. A preregistered annotation scheme will instead identify relations such as:

- an epistemic expression introducing proposition p ;
- later discourse maintaining or reasoning within the p -possibility;
- evidence confirming, weakening, or excluding p ; and
- alternative expressions that realize comparable discourse functions.

A manually annotated sample will establish whether these relations can be identified reliably before any automatic extraction is attempted. The audit will estimate their frequency, conditional distribution, redundancy, and lexical or genre dependence. Its purpose is to turn a semantic hypothesis into a concrete hypothesis about learnable evidence.

3.3 Controlled learning experiment

Small language models will be trained from scratch under a fixed architecture, tokenizer, token budget, optimization protocol, and seed plan. Only the theoretically relevant input relation will vary:

- **Preserve:** retain the target relation;
- **Remove:** remove its direct instances while matching the training budget;
- **Structure-disrupt:** retain words and local statistics as far as possible while breaking the correct cross-sentence correspondence; and
- **Restore:** add the correspondence back to the disrupted baseline.

Evaluation will test maintenance, downstream use, and evidence-sensitive withdrawal of p on held-out lexical items, discourse templates, entities, and distances. Model probabilities will not be equated with human reaction times. The comparison concerns the direction and structure of condition effects, not a claim that the two systems implement the same mechanism.

Two computational paths keep the project feasible. **Path A**, a controlled miniature corpus, provides a low-cost test with known input statistics and can establish whether the intervention and evaluation are capable of producing an interpretable learning difference. **Path B**, a matched intervention on developmentally plausible naturalistic data, provides the stronger natural-language result but requires reliable annotation and substantially greater training and confound control. Path B will be attempted only if Path A and the corpus audit pass their prespecified gates.

4. Scope, inference, and stopping rules

The core project is limited to English, one primary contrast centred on *might p*, one adult behavioural experiment, one input audit, and one controlled training comparison. Pretrained-model evaluation is used only to calibrate the measurement and provide baselines. A child experiment, cross-linguistic work, activation steering, and internal-representation analysis are not required for completion. Recent work relating children's modal comprehension to non-verbal modal concepts (Leahy, 2026) makes the developmental connection theoretically valuable, but it is best treated here as a future extension rather than a third empirical study.

The project will stop or narrow at three points. First, if the human contrast disappears after controlling lexical repetition, subjective probability, general uncertainty, and discourse coherence, the thesis will not interpret it as epistemic-possibility accessibility. Second, if the candidate input relation cannot be annotated reliably or manipulated without altering unrelated distributions, the natural-corpus causal claim will be dropped and the computational result restricted to the miniature corpus. Third, if simple lexical or frequency baselines account for the model results, the thesis will not claim acquisition of an abstract modal-discourse behaviour.

Any causal conclusion will be learner-relative: a matched input intervention can show that a relation contributes to the behaviour of a specified architecture under a specified training regime. It cannot by itself show that children use the same cue, that the cue is necessary for human acquisition, or that the model internally implements the formal theory.

5. Expected contribution

The thesis aims to contribute a single connected evidence chain: a carefully delimited human behavioural effect, an explicit account of the linguistic evidence that could support it, and a controlled test of whether that evidence is sufficient for a neural learner to acquire and generalize the behaviour. The broader contribution is methodological: formal semantic distinctions define what must be learned; corpus evidence motivates a learning hypothesis; and controlled training results feed back on how the target behaviour and its possible sources should be decomposed.

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